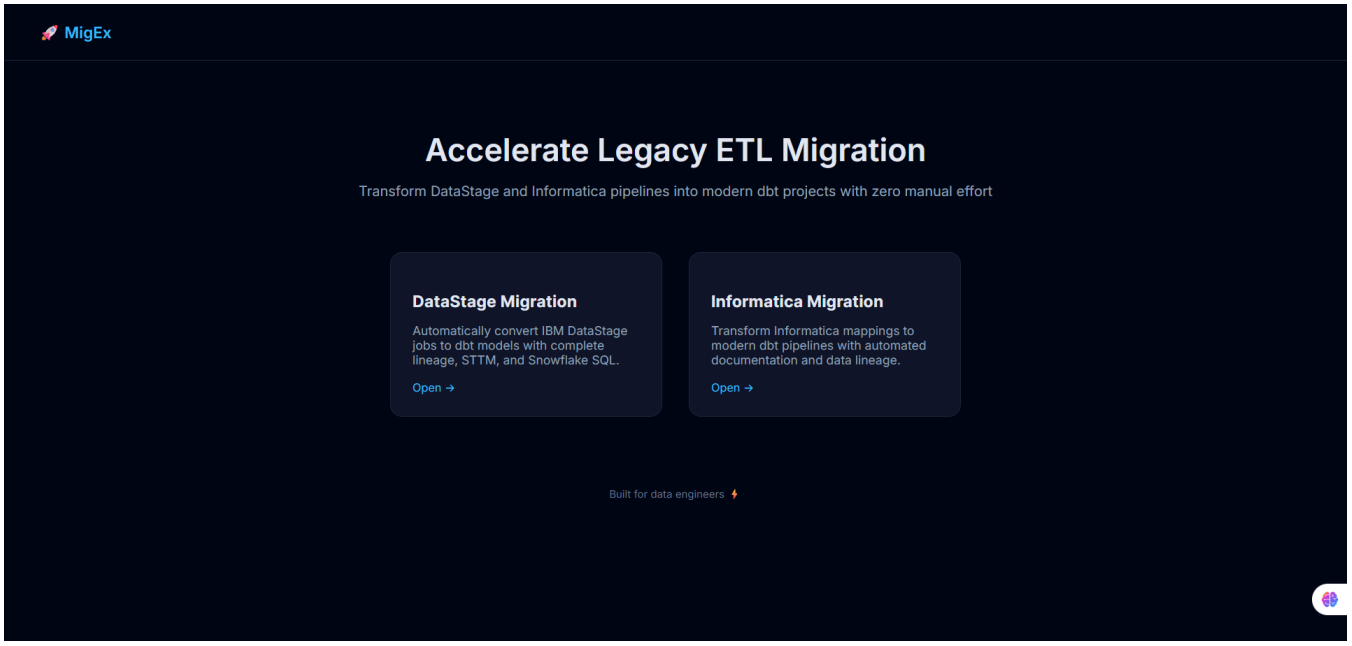
DataModernizer Engine PLATFORM v1.0.0

Complete Technical & Business Documentation

Version: 1.0.0 | **Classification**: Internal Use | **Audience**: All Stakeholders



EXECUTIVE OVERVIEW

DataModernizer Engine is an enterprise-grade ETL modernization platform that automates the transformation of legacy Informatica PowerCenter and DataStage DSX workflows into cloud-native Snowflake data architectures.

Business Impact at a Glance

Metric	Value	Impact
Time Reduction	TBD(based on usecase complexity)	TBD(based on usecase complexity)
Cost Savings	TBD(based on usecase complexity)	TBD(based on usecase complexity)
Automation Rate	90%+	Minimal manual work
Error Reduction	95% fewer issues	Confidence scoring on all transformations
Scalability	Multiple files/batch	Parallel processing via Celery

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PROJECT OVERVIEW

What It Does

DataModernizer Engine takes legacy ETL definitions and generates production-ready cloud data architectures:

Input → Analysis → Generation → Output

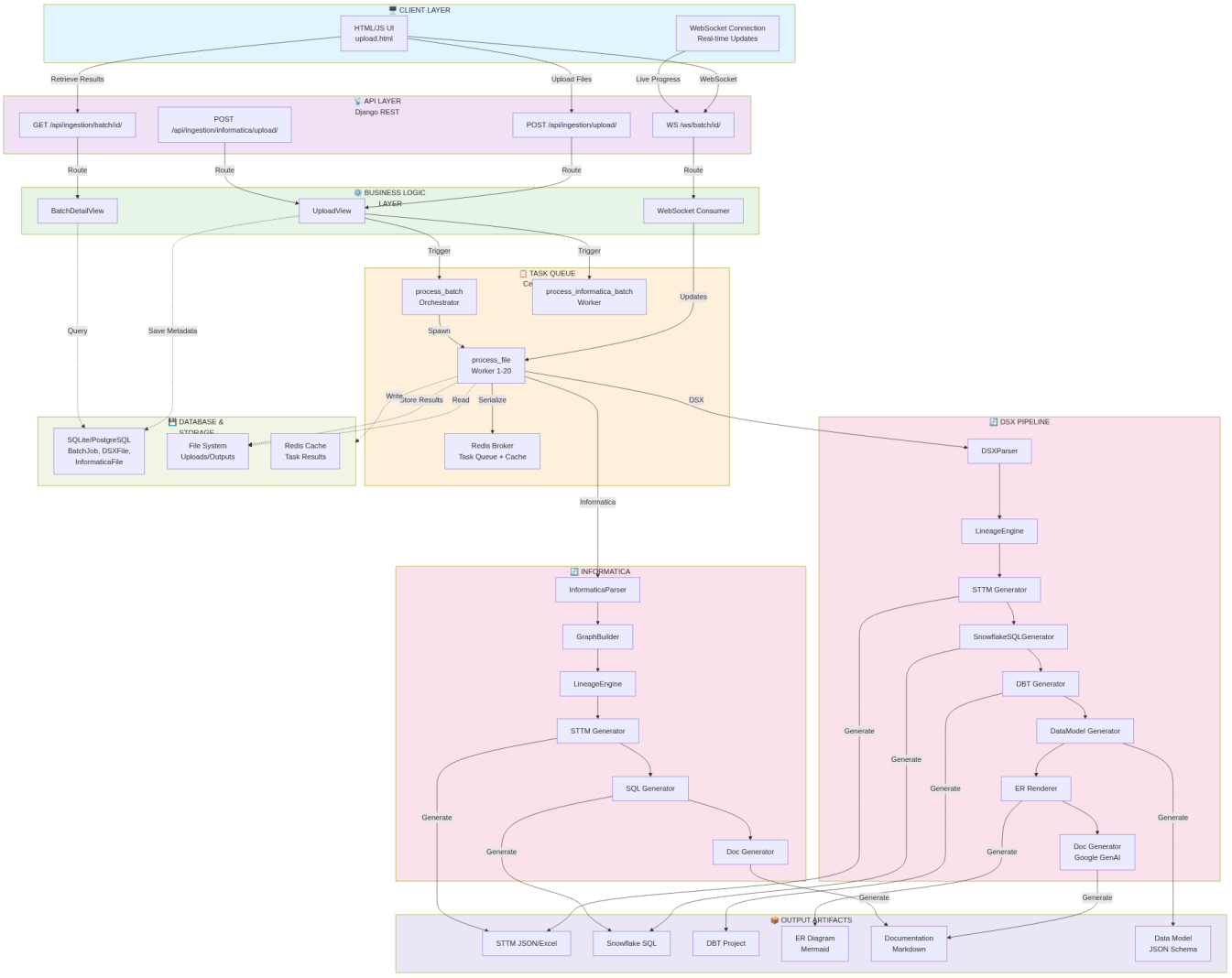
```
Legacy ETL Files          |         Processing Engines      |  Production Artifacts
                          |                                 |
• DataStage (.dsx)        |  ┌─ Parser                      |  • STTM (Mapping Doc)
• Informatica (.xml)      |  ├─ LineageEngine               |  • Snowflake SQL
• Multiple files/batch    |  ├─ STTM Generator              |  • DBT Project
  (parallel)              |  ├─ SQL Generator               |  • ER Diagram
                          |  ├─ DBT Generator               |  • Data Model JSON
                          |  ├─ Doc Generator               |  • Auto Documentation
                          |  └─ LLM Enhancement             |  • Processing Metrics
```

Version 1.0.0 Status

Component	Status	Quality
DataStage DSX Support	✅ Complete	Production-Ready
STTM Generation	✅ Complete	Production-Ready
Snowflake SQL	✅ Complete	Production-Ready
DBT Integration	✅ Complete	Production-Ready
ER Diagrams	✅ Complete	Production-Ready
Real-time Dashboard	✅ Complete	Production-Ready
Informatica Support	🔄 80% Complete	Phase 2 In Progress
Informatica SQL Gen	🔄 70% Complete	Phase 2 In Progress

ARCHITECTURE & COMPONENTS

System Architecture



Technology Stack

Layer	Technology	Version	Purpose
Web Framework	Django	6.0.3	REST API + Admin
Task Queue	Celery	Latest	Async processing
Message Broker	Redis	Latest	Task queue + WebSocket
ASGI Server	Daphne	Latest	WebSocket support
Database	SQLite/PostgreSQL	Any	Metadata storage
ORM	Django ORM	Built-in	DB abstraction
REST API	DRF (Django REST)	Latest	API serialization
Real-time	Django Channels	Latest	WebSocket messaging
LLM Integration	Google GenAI	Latest	Documentation/Enhancement
Data Processing	Pandas	Latest	Transformations
Excel Export	OpenPyXL	Latest	STTM export
Template Rendering	Jinja2	Latest	SQL generation

FEATURES & CAPABILITIES

1. Multi-Format ETL Input

DataStage DSX (✅ COMPLETE)

- **Format**: XML-based ETL metadata
- **Parsing**: Regex-based extraction
- **Elements Captured**:
  - Job definitions
  - Stage types and configurations
  - Column definitions with derivation logic
  - Data flow links between stages

Informatica PowerCenter (🔄 80% COMPLETE)

- **Format**: XML mapping metadata
- **Parsing**: XML ElementTree
- **Elements Extracted**:
  - Mapping definitions
  - Transformation instances (Expression, Lookup, Join, etc.)
  - Source/Target definitions
  - Connection string metadata

2. Source-to-Target Mapping (STTM)

STTM is the **data lineage backbone** - a structured mapping of every source column through transformations to target columns.

STTM Structure:

```
{
  "source": "CUSTOMER_ID, ACCOUNT_NUMBER",
  "target": "hk_customer_account",
  "transformation": "MD5(UPPER(CUSTOMER_ID) || ACCOUNT_NUMBER)",
  "stage": "Hash_Key_Calculation",
  "type": "DERIVED",
  "confidence": 95,
  "incomplete": false
}
```

Transformation Types:

- **DIRECT**: 1:1 column mapping, no transformation
- **SYSTEM**: Auto-generated (load_date, record_source)
- **DERIVED**: Transformed from multiple sources

- **LOOKUP**: Join with reference table
- **JOIN**: Multi-table join operation

  Quality Scoring Algorithm:

```
confidence = 100
if incomplete:
    confidence -= 40  # Missing logic
if type == "DERIVED":
    confidence -= 10  # Complexity
if type == "SYSTEM":
    confidence -= 20  # Auto-generated
if multiple_sources:
    confidence -= 10
confidence = max(0, confidence)
```

Output Formats:

- Excel (.xlsx) for business users
- HTML table for web display

3. Snowflake SQL Generation (DBT Version)

Generates production-ready Snowflake SQL following Data Vault patterns.

Staging Layer (Raw data import):

```
{{config(materialized='view')}}

SELECT
    MD5(CUSTOMER_ID) as hk_customer_id,
    CUSTOMER_ID,
    LOAD_DATE,
    'ETL_SOURCE' as record_source
FROM RAW_CUSTOMER_TABLE
```

Integration Layer (Hub/Link/Satellite):

```
-- Hub Table
CREATE OR REPLACE TABLE hub_customer AS
SELECT DISTINCT
    MD5(TRIM(UPPER(customer_id))) as hk_customer_id,
    customer_id,
    CURRENT_TIMESTAMP as load_date,
    'SYSTEM_SOURCE' as record_source
FROM stg_customer
WHERE customer_id IS NOT NULL;

-- Satellite (SCD Type 2)
CREATE OR REPLACE TABLE sat_customer_details AS
SELECT
    hk_customer_id,
```

```
    load_date,
    MD5(CONCAT_WS('||', customer_name, customer_address)) as hash_diff,
    customer_name,
    customer_address,
    customer_status,
    CURRENT_TIMESTAMP as end_date
FROM int_customer;
```

Features:

- Satellite hash diff for change detection
- SCD Type 2 history tracking
- Incremental load patterns
- Snowflake-native functions (VARIANT, SEMI-STRUCTURED)

4. DBT Project Generation

Auto-generates complete dbt project scaffolding ready for deployment.

Directory Structure:

```
models/
├── staging/
|   ├── stg_customer.sql
|   ├── stg_account.sql
|   └── stg_transaction.sql
├── intermediate/
|   ├── int_customer_enriched.sql
|   ├── int_account_aggregates.sql
|   └── int_transaction_facts.sql
├── marts/
|   ├── dim_customer.sql
|   ├── dim_account.sql
|   └── fct_transactions.sql
├── schema.yml       # Documentation + tests
├── dbt_project.yml  # Project configuration
```

Generated SQL Example:

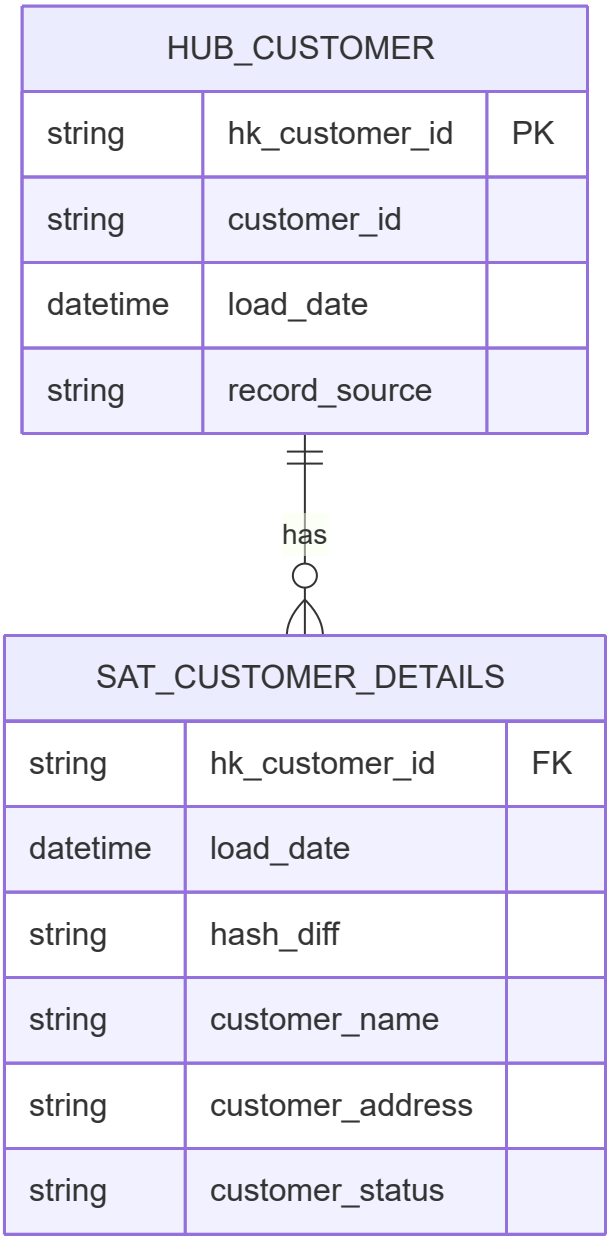
```
{{config(materialized='table', tags=['daily'], unique_id='dim_customer')}}

SELECT
    {{ dbt_utils.surrogate_key(['hk_customer_id']) }} as customer_key,
    hk_customer_id,
    customer_id,
    customer_name,
    load_date,
    current_timestamp as updated_at
FROM {{ ref('int_customer_enriched') }}
WHERE status = 'ACTIVE'
```

5. Data Model & ER Diagrams

Automatically generates Entity-Relationship Diagrams in Mermaid format.

Generated ERD Example:



Features:

- Automatic table relationship detection
- Primary/Foreign key visualization
- Cardinality mapping
- Mermaid syntax rendering in browser
- Interactive diagram exploration

6. Real-Time Processing Dashboard

WebSocket-powered live monitoring with comprehensive UI.

Dashboard Components:

Batch Statistics (Live Updates):

- Total files in batch
- ✅ Completed count
- ⏳ In-progress count
- ❌ Failed count

  File Display (Single File at a Time):

- Filename and status
- Progress bar (0-100%)
- Processing time per stage
- Navigation (Previous/Next buttons)

  Tab-Based Content View:

- **STTM Tab**: Interactive mapping table
- **SQL Tab**: Generated Snowflake DDL with syntax highlighting
- **Data Model Tab**: JSON schema viewer
- **ER Tab**: Mermaid diagram rendering
- **Docs Tab**: Auto-generated documentation
- **DBT Tab**: Project file tree explorer

7. Parallel Processing with Celery

Leverages Celery + Redis for concurrent file processing.

Queue Architecture:

```
Redis Message Broker
  ├─ Task Queue: process_batch tasks
  ├─ Task Queue: process_file tasks (20 concurrent)
  ├─ Channel Layer: WebSocket messaging
  └─ Results Backend: Task status

Celery Worker Pool
  ├─ Worker 1-20 (Thread Pool)
  ├─ Error Handling: Retry on failure
  ├─ Timeout: 1 hour per file
  └─ Concurrency: Configurable (default 20)
```

Processing Sequence:

1. User uploads 10 DSX files
2. UploadView creates BatchJob + 10 DSXFile records

3. process_batch.delay(batch_id) queued
4. Celery picks up task, spawns 10 process_file tasks
5. Up to 20 files processed in parallel
6. WebSocket updates sent in real-time
7. Batch completion detected automatically

Informatica Pipeline (🔄 In Progress)

Status: 80% complete - Phase 2 Q2 2026

Current Implementation (✅ Complete):

1. InformaticaParser - XML parsing
2. Mapping extraction
3. Instance classification
4. Basic STTM generation

   In Progress (🔄 70% - Phase 2):

5. Graph building & visualization
6. Advanced lineage tracing
7. Transformation rule translation
8. Connection string mapping

   ⏳ Phase 2 Q3 - **To Be Determined**

DATABASE SCHEMA

Core Models

BatchJob

```
class BatchJob(models.Model):
    id: int (PK)
    status: CharField (PENDING, PROCESSING, COMPLETE, PARTIAL_FAIL)
    created_at: DateTime (auto timestamp)
    completed_at: DateTime (nullable, set when all files done)
```

DSXFile

```
class DSXFile(models.Model):
    id: int (PK)
    batch: ForeignKey(BatchJob, CASCADE)
    file: FileField (uploaded .dsx file)
    status: CharField (UPLOADED, PROCESSING, DONE, FAILED)

    # Outputs
```

```
    sttm_json: JSONField (mapping data)
    sttm_file: FileField (.json export)
    sttm_excel: FileField (.xlsx export)

    snowflake_sql: TextField (complete SQL)
    dbt_sql: TextField (dbt YAML)
    dbt_files: JSONField (file manifest)

    data_model: TextField (JSON schema)
    er_diagram: TextField (Mermaid syntax)

    documentation: TextField (Markdown)

    created_at: DateTime
    completed_at: DateTime (nullable)
```

InformaticaFile

```
class InformaticaFile(models.Model):
    id: int (PK)
    batch: ForeignKey(BatchJob, CASCADE)
    file: FileField (uploaded .xml mapping)
    status: CharField (PENDING, PROCESSING, DONE, FAILED)

    sttm_json: JSONField
    snowflake_sql: TextField
    documentation: TextField

    created_at: DateTime
    completed_at: DateTime (nullable)
```

Migration History

- **0001_initial.py**: Base schema
- **0002_dsxfile_extensions.py**: Added output fields
- **0003_informaticafile.py**: Informatica support
- **0006_batchjob_completed_at.py**: Timing metrics

API REFERENCE

Upload DSX Files

```
POST /api/ingestion/upload/

Request:
  Content-Type: multipart/form-data
  files: [File1.dsx, File2.dsx, ...]

Response (200):
```

```
{
  "batch_id": 42,
  "file_count": 2,
  "created_at": "2026-04-04T10:30:00Z"
}
```

Get Batch Results

```
GET /api/ingestion/batch/{batch_id}/

Response (200):
{
  "batch_id": 42,
  "batch_status": "COMPLETE",
  "batch_total_time_seconds": 4500.25,
  "file_count": 2,
  "files": [
    {
      "id": 101,
      "name": "customer_etl.dsx",
      "status": "DONE",
      "processing_time_seconds": 2250.15,
      "sttm": [...],
      "snowflake_sql": "...",
      "dbt_files": {...},
      "data_model": "{...}",
      "er_diagram": "...",
      "documentation": "..."
    }
  ]
}
```

Upload Informatica Files

```
POST /api/ingestion/informatica/upload/

Request:
  files: [Mapping1.xml, Mapping2.xml, ...]

Response (200):
{
  "batch_id": 43,
  "file_count": 2,
  "created_at": "2026-04-04T12:00:00Z"
}
```

WebSocket Real-Time Updates

```
Connection:
  ws://localhost:8001/ws/batch/{batch_id}/

Events Received:

  File Start:
  {
    "file_id": 101,
    "file_name": "customer_etl.dsx",
    "status": "PROCESSING",
    "step": "START"
  }

  Progress Update:
  {
    "file_id": 101,
    "step": "PARSING",
    "progress": 15
  }

  File Complete:
  {
    "file_id": 101,
    "status": "DONE",
    "step": "DONE",
    "processing_time_seconds": 2250.15,
    "sttm": [...],
    "snowflake_sql": "...",
    "dbt_files": {...}
  }

  Batch Complete:
  {
    "type": "BATCH_COMPLETE",
    "batch_status": "COMPLETE",
    "timestamp": "2026-04-04T11:45:00Z"
  }
```

INTEGRATION DETAILS

FIBO Compliance

FIBO (Financial Industry Business Ontology) alignment ensures regulatory compliance:

✅ **Data Lineage** (STTM):

- Complete source-to-target mapping
- Transformation documentation

- Audit trail of all changes

✅ **Metadata Management**:

- Comprehensive column mapping
- Business term definitions
- Data glossary generation

✅ **Compliance Reporting**:

- Processing audit trail
- File-level metrics
- Batch completion timestamps

✅ **Change Management**:

- Version control support
- Impact analysis ready
- Rollback metadata

Snowflake Integration

SQL Generation Patterns:

- Data Vault 1.0.0 Hub/Link/Satellite
- Incremental loading
- Snowflake native functions

Deployment Ready:

- Production-grade SQL
- Materialization strategies
- Performance optimization
- Cost optimization patterns

dbt Integration

Project Scaffolding:

- Layered model structure
- Auto-generated documentation
- Schema YAML
- Macro support for reusability
- Source definitions

ROADMAP

Phase 1: ✅ COMPLETE (Current)

- [✅] DataStage DSX parsing
- [✅] STTM generation
- [✅] Snowflake SQL generation
- [✅] DBT project scaffolding
- [✅] ER diagram generation
- [✅] Real-time dashboard
- [✅] Parallel processing
- [✅] WebSocket live updates

Phase 2: 🔄 IN PROGRESS (Q3 2026)

- TBD

System Requirements

Minimum: 4GB RAM, 2 CPU, 50GB disk
Recommended: 32GB RAM, 8 CPU, 500GB SSD
Scale: 100+ files/batch, 20 concurrent tasks